

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3				Indicative content: Origin of the Universe Both provide evidence to support the Big Bang when the Universe originated from a point. CMBR At the time of the Big Bang high energy short wavelength em waves were produced (gamma rays). Over time as space expanded, these em waves also stretched / became longer forming microwaves. The CMBR fills all of space with (nearly) uniform intensity in all directions. Cosmological red shift Evidence for the expansion of the Universe since the radiation was emitted. As the space between galaxies stretches any light moving through space will also be stretched and its wavelength will increase i.e. be red shifted. This is revealed in the study of spectra since absorption lines have increased wavelengths. 5-6 marks Detailed description/explanation of cosmological red shift, CMBR and how they support the origin of the Universe. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Detailed description/explanation of cosmological red shift or CMBR OR limited description/explanation of cosmological red shift and CMBR. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Limited description/explanation of cosmological red shift or CMBR or reference to how they support the origin of the Universe. <i>There is a basic line of reasoning which is not coherent, largely irrelevant,</i>	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				<i>supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i>						
				0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 3 total	6	0	0	6	0	0